



Diode Breakdown Test Procedure

CLS Test Procedure – 1.7.32.1 Rev. 0

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REVISION HISTORY

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1.0 PURPOSE

This document outlines the test procedure required for the diode breakdown test.

1.1 BACKGROUND

Regular preventive maintenance will prevent catastrophic failure of components of the modulators.

1.2 QUALIFICATIONS

Electronic Technician, Technologist or Engineer.

1.3 EQUIPMENT AND MATERIALS

Universal-Voltronics portable High Voltage Power Supply.

H.V. Gloves

$\frac{3}{4}$ inch open end wrench

5/16 inch common tip screwdriver

2.0 PROCEDURE FOR TESTING FOR DIODE BREAKDOWN

Lock and tag out the modulator AC at panels DPNL 1021-01 and DPNL 1022-02.

Short out the PFN using the shorting rods provided.

Remove the diode stack from the modulator. This is accomplished by removing the cables from both ends of the stack using the screwdriver and removing the nuts using the $\frac{3}{4}$ inch open end wrench.

Place the stack on a non-conducting bench or table.

Verify that the HV Gloves have not exceeded the "twelve month shelf" life or the "six months since used" life. If the gloves have exceeded the life time they must be retested before use.

Put on the H.V. gloves.

The diodes are forward biased when the negative lead is on the base of the diode and the positive lead is on the curved part.

Connect the shielded (positive) lead of the high voltage power supply to the top plate of the diode stack and connect the black (negative) lead to the second diode plate so that the diodes are reverse biased.

On the Universal-Voltronics portable High Voltage Power Supply:

Position the polarity switch in the positive position.

Turn the "H.V. Control" fully counter clockwise.

Switch the "ON-OFF" switch to the ON position.

Depress the "High Voltage On" (black) button.

Turn the "High Voltage Control" to 2.5 kV.

The milliampere meter reads zero if the diode is good. The maximum permissible amount of reverse current is 10uA.

Turn the power supply off by depressing the "High Voltage Off" (red) button.

Set the On-Off switch to off.

Touch the negative lead to the positive lead to ensure high voltage is discharged.

Move the shielded (positive) lead to the second plate of the diode stack.

Move the black (negative) lead to the third plate of the diode stack.

Turn the "H.V. Control" fully counter clockwise.

Switch the "On-Off" switch to the On position.

Depress the "High Voltage On" (black) button.

Turn the "High Voltage Control" to 2.5 kV.

Ensure that the milliampere meter reads zero. The maximum permissible amount of reverse current is 10uA.

Repeat for every diode in the stack.

Replace any diode that shows more than 10uA of current on the milliampere meter.

Replace the diode stack into the modulator from which it came.

Continue with the next diode stack and until all of the diodes have been tested.

UDE2.5

Electrical Specifications (at 25°C unless noted)					Maximum Ratings				
Type	PIV	Maximum Forward Voltage Drop	Maximum Leakage Current @ PIV	Maximum Reverse Recovery Time	Maximum Average D.C. Output Current			Non-Repetitive Sinusoidal Surge (8.3ms) T _C = 100°C	Maximum Reverse Transient Energy Absorption
					T _C = 75°C Air	T _C = 60°C Air with Extender Plate**	T _C = 50°C Oil		
Standard Recovery	UDE-2.5	2.5	5V @ 3.00A	10	‡ 6.00	7.00	7.70	200	8
	UDB-2.5	2.5	4V @ 1.50A	5	3.00	3.75	4.25	100	4
	UDE-5	5	10V @ 2.20A	10	‡ 4.50	5.00	5.50	200	14
	UDB-5	5	8V @ 1.00A	5	2.00	2.50	2.75	100	8
	UDA-5	5	8V @ 0.82A	2	1.65	2.00	2.20	30	1.5
	UDB-7.5	7.5	12V @ 0.70A	5	1.33	1.65	2.00	100	12
	UDA 7.5	7.5	12V @ 0.60A	2	1.25	1.55	1.75	30	2.5
	UDA-10	10	16V @ 0.50A	2	1.00	1.25	1.40	30	3
Fast Recovery	UDA-15	15	25V @ 0.33A	2	0.67	0.80	0.90	30	5
	UDF-2.5	2.5	6V @ 2.20A	10	4.50	5.00	5.30	150	8
	UDD-2.5	2.5	6V @ 1.20A	5	2.25	2.80	3.30	80	4
	UDF-5	5	11V @ 1.60A	10	3.30	4.00	4.40	150	14
	UDD-5	5	11V @ 0.75A	5	1.50	1.85	2.00	80	8
	UDC-5	5	10V @ 0.70A	2	1.20	1.50	1.70	25	1.5
	UDD-7.5	7.5	17V @ 0.50A	5	1.00	1.25	1.50	80	12
	UDC-7.5	7.5	15V @ 0.50A	2	0.90	1.10	1.25	25	2.5
	UDC-10	10	20V @ 0.37A	2	0.75	0.90	1.00	25	3
	UDC-15	15	30V @ 0.25A	2	0.50	0.60	0.70	25	5

*Measured in a reverse recovery circuit switching from 1.0A forward to 1.0A reverse current recovering to 0.5A.

†Measured in a reverse recovery circuit switching from 0.5A forward to 1.0A reverse current recovering to 0.25A.

**These ratings are based on using "extender plates" that provide additional surface area to radiate heat. Because of possible corona effects caused by scratches on these plates, extreme care is necessary in their handling and they are not recommended where the working voltage exceeds 7.5KV/module. They should be carefully polished prior to installation.

‡These ratings are based on T_C = 100°C.